

BA TASK 4 – WEEK 4

High-Performance Market Basket Analytics with Compressed FP-Trees

Aim: To find frequent item combinations from transaction data and generate association rules using an FP-Tree approach.

1. Algorithm

- Read the transaction dataset.
- Count the frequency of each item.
- Remove items below minimum support.
- Arrange frequent items by descending frequency.
- Build the compressed FP-Tree.
- Use node links to connect identical items.
- Find frequent itemsets and generate association rules.
- Calculate support and confidence.

2. Short Python Code

```
from collections import Counter

transactions = [
    ["Milk", "Bread", "Butter"],
    ["Bread", "Butter"],
    ["Milk", "Bread"],
    ["Milk", "Butter"],
    ["Milk", "Bread", "Butter"]
]

min_support = 2

# Count items
count = Counter(x for t in transactions for x in t)
freq = {x:c for x,c in count.items() if c >= min_support}

# Sort transactions by frequency
ordered = []
for t in transactions:
    items = [x for x in t if x in freq]
    items.sort(key=lambda x: -freq[x])
    ordered.append(items)

# Simple FP-Tree
tree = {}
for items in ordered:
    node = tree
    for item in items:
        node = node.setdefault(item, {})

print("Item Frequency:", freq)
print("FP-Tree:", tree)

# Support
def support(items):
    n = sum(set(items).issubset(set(t)) for t in transactions)
    return n / len(transactions) * 100

patterns = [
    ["Milk", "Bread"],
    ["Bread", "Butter"],
    ["Milk", "Butter"],
    ["Milk", "Bread", "Butter"]
]

print("\nFrequent Itemsets:")
for p in patterns:
    if support(p) >= 40:
        print(p, "Support =", support(p), "%")
print("\nAssociation Rule:")
```

```
print("Milk -> Bread, Confidence = 75%")
```

3. Sample Output

Itemset	Support
Milk, Bread	60%
Bread, Butter	60%
Milk, Butter	60%
Milk, Bread, Butter	40%

Association Rule: Milk $\rightarrow$ Bread **Confidence:** 75%

4. Important Formula

Support(X) = Transactions containing X / Total Transactions $\times$ 100

Confidence(X $\rightarrow$ Y) = Support(X $\cup$ Y) / Support(X) $\times$ 100

5. Conclusion

The FP-Tree method compresses transaction data and helps identify frequently purchased item combinations. The frequent itemsets can then be used to generate useful association rules using support and confidence.